Slot Design

The design of a slot game must meet three essential requirements. First, it should enable seamless back-end implementation. Second, it must allow for rich, flexible configuration. And third, it needs to deliver a compelling payout profile.

Let's be direct: if there were no money involved in slot machines, all the other elements—the graphics, animations, sounds, and overall presentation—would not be enough to make up for the core appeal of gambling. The payout mechanics are what truly drive player engagement, and everything else builds on that foundation.

Commonly used terms (glossary):

- **Random Variable:** From a mathematical standpoint, every slot machine is a Random Variable. What defines a Random Variable is its parameters, with the distribution being the most important—since the distribution contains all other parameters and they can be derived from it.

- **Event:** Any bonus, feature, or other winning combination in a game is called an event. Events alter the behavior of the Random Variable, changing how the game performs.

1. Physical and Combinatorial Requirements (Scenarios and Variability)

Every event, every possible combination of events, and every edge case must be logically executable. In other words, the designer should have mentally walked through all possible scenarios that can arise from the game's specification—imagining how they would actually play out visually—so that the logic behind those events can be properly implemented in both the back-end and front-end.

These events must also align with parameters established at the design level, such as screen size and basic payout strategy, to ensure they behave as intended. For a deeper dive into this topic, see the detailed description [HERE].

2. Mathematical and Statistical Requirements (Laws)

All quantitatively defined events must conform to the mathematical and statistical framework established by the parameters of a Random Variable. This ensures that all regulatory requirements are met—for example, that a 100% Return to Player (RTP) is not exceeded.

Any controllable Random Variable—such as a slot machine—can be modeled with a desired target distribution. However, to maintain control over it, all events must be properly parameterized at the design stage, providing the mathematical model with enough variability to work effectively.

Many of these requirements cannot be fully predicted during design, but a solid foundation must be laid here. For a more detailed discussion, see the full description [HERE].

3. Originality and Natural Build on Base Mechanics (Simplicity)

Is the event intuitive enough for a player to grasp? Can it be explained easily to someone with no technical background, so they immediately understand what it is and what to expect from it?

If you were a player, not a designer, would you be genuinely excited to encounter this event during gameplay?

Most importantly: Does this event feel like a natural extension of the core payout mechanics—whether lines, ways, clusters, scatters, or others—or does it feel forced and disconnected from the base game?

For more on why simplicity is essential to this requirement, see the detailed discussion [HERE].

Combinatorial Requirements

1. Physical and Combinatorial Requirements (Scenarios and Variability)

- **PRIORITIATION**

A wild, or substitute symbol, can stand in for other symbols to help form winning combinations. For example, imagine a game where the wild symbol W substitutes for symbols A and B. If the reels stop with the sequence {W, W, A} on a particular payline, this line qualifies for both a three-of-a-kind win with A (A-A-A) and a two-of-a-kind win with B (B-B).

While both wins could be paid, standard practice in slot games is to pay only the highest win per payline. This is known as **prioritization**—ordering the pay rules by value and awarding only the top one.

Wilds Pitfall

A wild symbol is defined solely by its ability to substitute for other symbols. Whenever a designer assigns an explicit payout to a wild symbol in the paytable, it breaks the mechanics and introduces a synthetic error. By definition, a wild's payout is implicit—it pays as the symbol it substitutes for in any given winning combination, but only as one of them per win.

If a winning combination could be interpreted in multiple ways with different payouts, the highest win takes precedence, following the prioritization rule. In the edge case where a winning combination consists entirely of wild symbols, they collectively substitute for the highest paying symbol on the paytable.

OVERFLOW

Consider the following feature specification, inspired by a similar mechanic in the game *Paddy Power: Golden Leprechaun: Knock! Knock! Knock!*:

"From 2 to 4 wilds appear randomly on the reels, each of which can be cloned with 0 to 4 new wilds."

The game screen has 5 reels and 3 rows—a total of 15 positions. If all feature parameters are maximized, we would need to place 16 wilds on screen: 4 original wilds, each producing 4 clones. That's one wild too many. Where would they go?

While the solution might seem straightforward, the semantics of wild symbols require careful consideration. The language used in the specification often finds its way into the game's help text, which players will read and rely on. Therefore, any ambiguity must be addressed at the design stage.

Possible Solutions

- One approach is to adjust the parameters: for example, reduce the original wilds to a range of 1 to 3, and increase the clones to a range of 0 to 5. This keeps the maximum potential at 15 wilds.
- Another solution allows wilds to overlap. If a new wild lands on an existing wild position, the multiplier of that wild increases by 1. Wilds start with a multiplier of ×1. Each multiplier applies only to the payline it participates in, and if multiple wilds with multipliers greater than ×1 appear in the same win, their multipliers are added together (the aggregation function is addition).

A More Complex Case: Scatters

Now consider what happens if the symbols being cloned are scatters rather than wilds. Scatters are special symbols with their own payout mechanics; they can trigger unique events or function independently of the base payout strategy. Scatters often block normal line pay calculations.

If the feature could theoretically produce 16 scatters—but the screen only holds 15—we face the same overflow problem. However, unlike wilds, scatters cannot simply overlap or merge without breaking their intended behavior. A radical solution might be to make scatters function as wilds in this context, but that introduces new questions: Would this feel natural to players? Would it create unintended side effects?

In an abstract game, we cannot know. But the key takeaway is this: such edge cases must be resolved at the design level. Designers should mentally simulate every scenario and make decisions that enhance the game experience—not introduce synthetic mechanics that undermine it.

VARIABILITY

Consider the following feature specification, inspired by a similar mechanic in the game *FanDuel: Classic Invaders*:

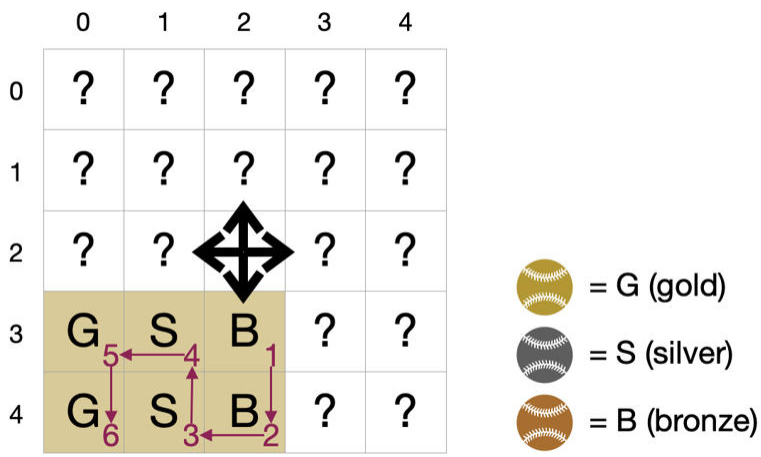
When the feature is triggered, a new screen appears showing a 5x5 grid of hidden tiles. The traversal begins at the center tile. From there, each step may only move to an adjacent tile that shares a common side and has not been visited before.

The grid contains 24 hidden symbols, plus one symbol at the starting position that is revealed immediately—making 25 total. There are exactly three symbol types: Gold, Silver, and Bronze, each with its own payout value.

The feature ends the first time the player collects three symbols of the same type during the traversal. At that moment, the corresponding prize for that symbol type is awarded.

Where Are the Problems?

Let's label the three symbol types for clarity: Bronze = B, Silver = S, and Gold = G. Now consider the following traversal path:



Let's examine the traversal path from the grid above. We have a path of length 6, and we find ourselves at a dead-end cell. According to the movement rules, we cannot continue—yet we have not collected three symbols of any single type. The feature cannot end, and no win is awarded. This is a problem.

Why This Happens

By the pigeonhole principle, any path of length 7 must contain at least three symbols of the same type. That would satisfy the end condition automatically. For paths shorter than 7, the feature ends the first time three matching symbols appear along the path.

But what about paths of length 6 that reach a dead end? In this case, the player is stuck with no way to continue and no win—clearly an unacceptable outcome.

Possible Solutions

We have two options:

1. **Populate the board strategically** so that every possible path of length 6 contains at least three identical symbols. However, this approach introduces a new problem.

2. **Change the movement rules** to allow more flexibility in traversal.

The Problem with Strategic Population

If we algorithmically guarantee that every path contains a win, we face regulatory challenges. After the bonus ends, the entire grid is revealed to the player. To comply with requirements, the board must contain at least one path leading to a gold win and at least one path leading to a silver win.

If we pre-populate the board to meet these conditions, the probability of hitting a gold path becomes no less than 1 in 8. Players could potentially intercept this pattern, making the game too predictable and restrictive. Any algorithm ensuring these conditions would need to account for all possible paths and select among them—without backtracking to validate the board's composition.

A Better Solution: Modified Traversal Rules

A cleaner approach is to change the traversal rules. For example, allow movement from one cell to another if they either:

- Share a common side, or
- Are at opposite ends of the same row or column (as long as the destination is unvisited)

This creates more path possibilities and reduces the chance of dead ends.

With this new rule, we can populate the board using a randomized, probability-based assignment for each symbol and cell. Then we run a backtracking algorithm to validate that the board contains at least one winning path of each type. The backtracking ensures compliance without making the outcomes predictable, while the randomization preserves the necessary variability.

But We're Not Done Yet

Most likely, the designer will want to allocate a specific portion of the game's total RTP to this feature—let's say 30%. They may also want the bonus to be available for purchase (a Bonus Buy option).

However, the feature as described has only three possible exit outcomes: Gold, Silver, or Bronze. This creates two problems:

1. **RTP Configuration Becomes Difficult**
   We cannot simply adjust the symbol distribution on the board. Any imbalance will cause the backtracking validation algorithm to fail, rejecting boards and slowing down the process significantly. Fine-tuning the RTP becomes a technical challenge.

2. **Bonus Buy Loses Appeal**
   With only three possible payouts, the feature offers too little variability to be interesting as a purchased bonus. A player who buys the feature knows they will win one of only three outcomes—hardly an exciting proposition.

The Solution: Layered Randomization

Once again, variability proves essential. The fix is to introduce an additional layer of randomness before the traversal begins.

For each path type, we can randomly select the actual payout from a range of possibilities. For example:

- **Gold path:** one payout selected from {x200, x500, x1000} stake multiplier
- **Silver path:** one payout selected from {x20, x50, x100} stake multiplier
- **Bronze path:** one payout selected from {x2, x5, x10} stake multiplier

This approach achieves two important goals:

- **Greater outcome variability**—the player no longer faces just three possible results, but multiple weighted possibilities within each tier.

- **RTP configurability**—by weighting the selection probabilities within each tier, we can precisely tune the feature's contribution to the overall RTP, hitting any target interval with ease.

The core mechanic remains intact, but the added layer of randomization transforms the feature from rigid and predictable to flexible and configurable.

This version clearly lays out the problem, explains why it matters, and presents the solution in a logical, compelling way.

DESCRIPTORS vs. PHYSICAL PLACEMENT

This case study is a specific instance of the broader variability topic, but it is extensive enough to warrant its own discussion. Let's examine a common feature specification:

"The game must have special symbols that appear in stacks of at least [x] symbols of the same type, and the hit rate for these special symbols must be Y (where Y < 1)."

This type of requirement appears explicitly or implicitly in almost every slot game specification. But it comes with hidden challenges.

The Problem with Physical Placement

If we physically place these stacked special symbols directly on the reels, we run into several issues:

- **Manual adjustment becomes tedious and error-prone**, especially when fine-tuning hit rates across many reels.

- **Context conflicts are hard to detect.** With games that have 10, 50, or even 100 reel sets, physically placed symbols might inadvertently be arranged so that certain special symbols never appear together—or never appear in combinations they should. These loopholes are difficult to spot and even harder to fix.

The Solution: Descriptors

The answer lies in moving from physical placement to symbolic descriptors. Instead of placing special symbols directly on the reels, we use descriptors that specify where on the reel a symbol could appear, without committing to a fixed physical position.

This approach unlocks powerful variability:

- Special symbols are no longer fixed in place, allowing them to appear in any context and combination as needed.

- The descriptor can carry additional information: stack size, screen position, behavior rules, and more.

- We can populate the reels richly with special symbols off-screen while keeping on-screen appearances controlled. This technique can increase near miss effects—but it can also make anticipation too high if not carefully managed. With great power comes great responsibility.

A Key Insight

Whenever a special symbol needs to be distributed across the reels, it is worth asking: *Does it need to be physically visible on the reels during the spin at all?*

Consider alternative approaches:

- Place symbols randomly after the reels have stopped, based on some logic.
- Place them before the spin but not during it.
- Use descriptors to generate placements dynamically.

Why This Matters

This shift from physical placement to descriptor-based placement is not just a technical detail. It is a design philosophy that prioritizes configurability and variability. The result is mechanics that are easier to develop, easier to tune, and—in many cases—lead to a better game feeling for the player.

The challenges introduced by this approach (overlapping symbols, blocking behavior, etc.) are solvable with experience. The core benefit—high configurability and true variability—is worth the effort.

Math Requirements

2. Mathematical and Statistical Requirements (Laws)

Any Random Variable over which we have control of the probability mass function can be modeled to achieve any target distribution. Every slot machine is precisely such a Random Variable. This means that any slot machine—regardless of its features—can be made mathematically equivalent in distribution to any other slot machine.

Now imagine that second machine has the perfect distribution—assuming such a thing exists. So what is the perfect distribution for a slot machine?

That is a non-trivial question. But we can begin to answer it by studying the best games on the market. And there are certainly many pure gems out there.

RTP Overflow and the Importance of Configurability

Once a game logic is implemented with maximum configurability—exporting the necessary parameters based on conditional probabilities and custom attributes in the configuration file—the RTP can sometimes reach hundreds or even thousands of percent above the target, even with the lowest possible parameters from the specification. This is why every feature needs a Plan B, and that plan must consider not just hit rates and mean values, but the feature's entire variability.

What the Design Team Must Know

The design team must be intimately familiar with three key elements:

1. **The configuration file** (now typically in JSON format)—easily accessible and human readable
2. **The PAR Sheet**—containing the mathematical foundation
3. **The back-end logic**—how features are triggered, whether forcibly or through conditional probabilities

This information is essential for diagnosing RTP overflow. The configuration file alone can reveal many root causes:

- Is the paytable too high, too low, or unfavorably distributed?
- Are the reels out of sync with the features that execute on them (stack sizes, symbol distribution)?

- Are there too few reel sets?
- Is there missing logic based on conditional probabilities?
- And many other issues that become visible on inspection.

Understanding the Parameters of a Random Variable

Every slot machine is, mathematically speaking, a discrete Random Variable. And every Random Variable is characterized by key parameters:

- **Frequency** (of predefined events)
- **Mean** (average value)
- **Median** (the middle value)
- **Variance** (spread of outcomes)
- **Volatility** (risk measure)
- **Distribution** (the complete picture)

Of these, **distribution is the most important**. It contains all other parameters within it—they can be derived using a moment-generating function or other methods. Each parameter must be understood deeply: what it means mathematically and how it manifests in the game feeling.

Practical Relationships

- The higher the average value of a feature, the lower its frequency will be.
- The closer the median and mean values are, the more consistent and less volatile the feature will be—and vice versa.
- The higher the variance of a bonus, the slower the RTP will converge during simulation, the more volatile the game, and the less likely Max Odds events become.
- The further skewness deviates from zero, the more synthetic the game feels—meaning probabilities are programmatically skewed away from a normal distribution. This can be a red flag, but in the hands of an experienced configurator, it can also be the key to a game's success.

The Trade-Off

We cannot have it all. A game with many features cannot have all of them be high-frequency, high-paying, and richly configured. Success is more likely with a smaller number of features—refined, easy for the player to understand, and perfectly configured.

Plan B: Designing for the Unexpected

It is not possible to predict with certainty whether a given set of events will fit into the specified parameters of a Random Variable—not for designers, not for programmers, not for configurators, and not even for Gauss, Fisher, Laplace, or Markov, on whose laws the entire gambling industry is built.

But what *can* and *must* be done is to design a Plan B for scenarios where RTP drastically exceeds its frame.

Plan B means this: events must be clearly defined and strongly parameterized, giving designers the ability to change parameters and conditions without disturbing the core mechanics at the conceptual level. These parameters should cover not just quantitative values, but purely mechanical ones as well:

- When and how is an event activated?
- By what steps does a quantitative parameter increase?
- In what scenarios is an event likely to occur, and in which is it locked?
- And so on.

Parameters should not be limited to hit rates and multiplier values. They must be more complexly thought out.

A well-designed feature contains a Plan B, and a Plan C, all the way to Plan Z—intuitively built into its structure. And the best features never even need to reach Plan B.

Natural Build-Up

3. Originality and Natural Build on Base Mechanics (Simplicity)

"*Complexity has nothing to do with intelligence, simplicity does.*"

You can google these words and find countless interpretations. Many people discuss what they were meant to say, and many disagree—each with their own arguments. Personally, I think they are as accurately stated as possible. But not everyone has the experience to fully grasp their meaning.

Let me offer my point of view—and I believe the author's as well. But first... pause for a moment. Think about these words. Interpret them for yourself.

What do you think they mean?

My Interpretation

What I believe the author is trying to say is this: our so-called "complexities" are shaped by the search for simplicity. To go from A to B, there is movement. First, there is tension. Second, there is motion. Finally, there is stillness—simply being there.

Anyone can make things complex. But can you make them simple and still solve the same problem? That is where intelligence comes in. Simplicity is the key. **Complexity alone can betray a lack of ideas.**

Do you agree?

Where Ideas Meet Mathematics

Any topic that challenges creativity, art, and originality is difficult to describe—especially when that creativity is built on a foundation of mathematics and combinatorics. But for this discussion, we have exact examples to guide us.

We will examine three slot games:

1. **The First** – A game that carries a brilliant, yet simple idea.
2. **The Second** – A game that tries to replicate that same brilliant idea and solve a problem, but fails and derails—yet in its failure, becomes something beautiful in its own right, though poorly configured.
3. **The Third** – A game that manages to replicate the idea of the first in a different way—and does it much better.

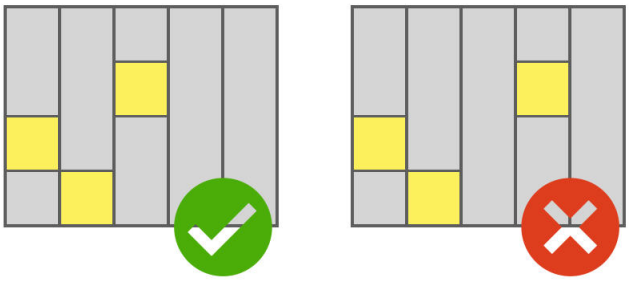
So let's begin.

WAYS (or also known as ALL WAYS)

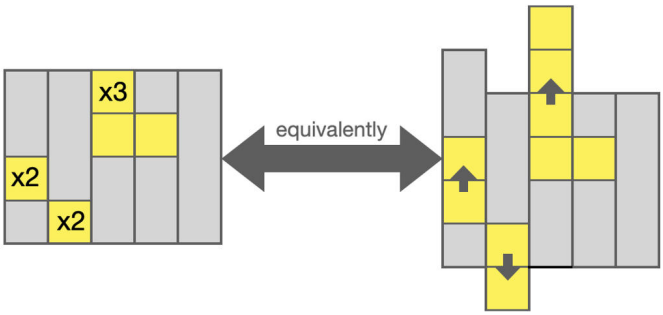
With WAYS payout strategy, **Wins are paid when matching symbols appear anywhere on adjacent reels from left to right starting from the leftmost reel**. This practically means that we include all possible simple lines and for a screen with size N reels and M rows we will have exactly M^N different payout matches. For example, for a screen of size Reels x Rows = 5 x 4, we will have total 1024 simple paylines.

MEGA WAYS

The **Megaways™** slot payout mechanic is the most played mechanic ever. It was developed by Big Time Gaming (BTG). They **patented** this innovative system, which allows for a dynamic number of paylines that can vary with each spin, providing a unique gaming experience. The mechanism has been **licensed** to various other game developers, allowing them to create their own Megaways slots using this popular format.



Special multipliers called **WAYS MULTIPLIERS** are defined particularly for this payout strategy. Those multipliers are decorators above the original symbols on which they are positioned. A way multiplier increases the **number of symbol occurrences** on the reel of the screen and in that way makes the symbol with the multiplier stronger in the ways calculation. Here is a visual example of a WAY MULTIPLIER and its interpretation:



While quite interesting and innovative, these multipliers create events that build on the mechanics synthetically and are not part of it. For this reason, you create events that are difficult to track and for this reason such games can hardly be certified in a region like Pennsylvania, as the max odds regulations (or at least those before 2024-11-12) restrict them.

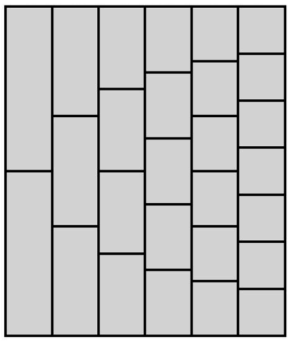
Key Features of Megaways™ Payout Strategy:

1. **Dynamic Paylines:**
   - Instead of fixed paylines, Megaways™ games use a "ways to win" system.
   - Each reel can display a different number of symbols (usually between 2 and 7 symbols per reel).
   - The total number of ways to win is calculated by multiplying the number of symbols on each reel. For example, if the reels display 3, 4, 5, 6, 5, and 4 symbols respectively, the total ways to win would be 3 × 4 × 5 × 6 × 5 × 4 = 7,200 ways.
2. **High Volatility:**
   - Megaways™ games are known for their high volatility, meaning wins may be less frequent but potentially larger when they occur.
   - The dynamic nature of the paylines and the cascading reels contribute to this volatility.
3. **Cascading Reels:**
   - Many Megaways™ games feature cascading or tumbling reels.
   - When a winning combination occurs, the winning symbols disappear, and new symbols fall into place, potentially creating additional wins in a single spin.

Advantages of Megaways™ Payout Strategy:

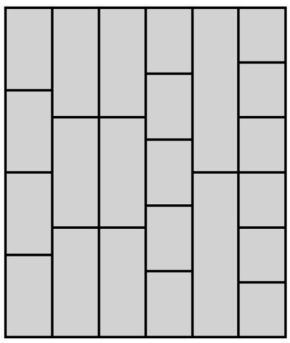
- **High Engagement:** The constantly changing number of ways to win keeps players engaged.
- **Massive Win Potential:** The high volatility and multipliers can lead to significant payouts.
- **Innovative Gameplay:** The dynamic nature of the mechanic sets it apart from traditional slot games.

2 x 3 x 4 x 5 x 6 x 7

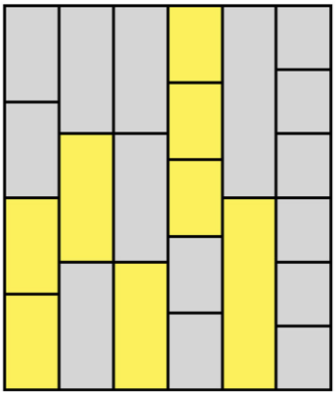


Total: 5040 ways

4 x 3 x 3 x 5 x 2 x 6



Total: 2160 ways

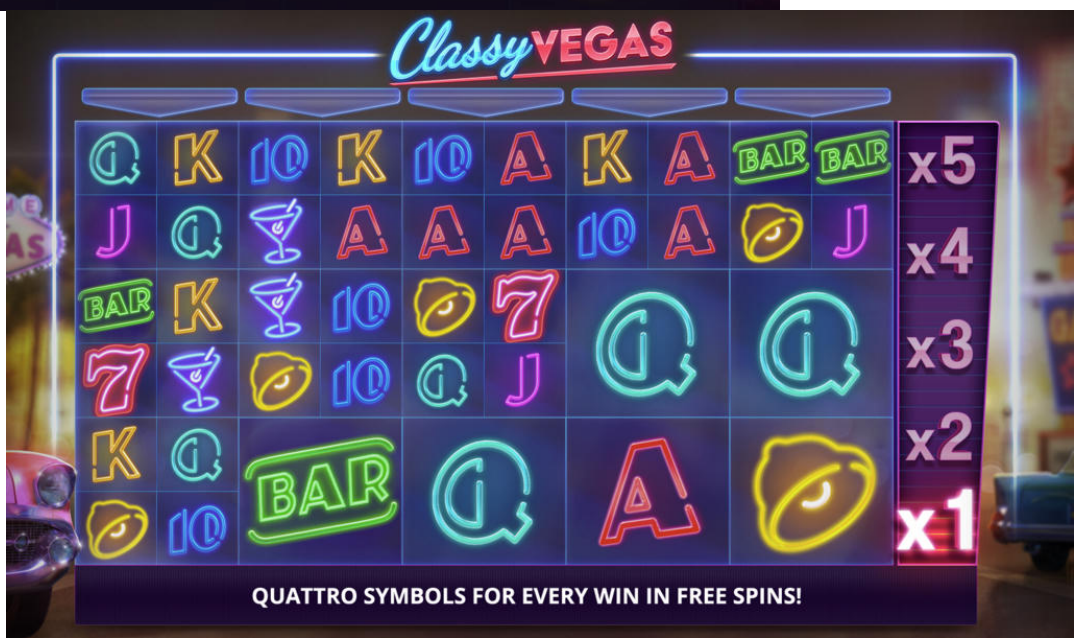


6 way win example of length 5

Notice how the WAY MULTIPLIERS that build on the natural WAY mechanics from the previous observed pay strategy now are already built into the MEGAWAYS mechanics and are part of it, rather than being synthetic events that build on it. This way, the megaways mechanics are not subject to any regulations. It's a brilliant idea that simultaneously injects a lot of variation, solves problems, and is beautiful. The idea behind this basic payout mechanic is to simply expand it with a feature functionality in it, which is a simple but not trivial idea.

The first game that we will observe is the Big Time Gaming's game **Bonanza Megaways** released in 2016. Bonanza was the **first-ever MegaWays slot**, introducing the groundbreaking mechanic to the world. It features a gold mining theme with 6 reels and up to 117,649 ways to win. The game includes free spins with unlimited win multipliers, making it highly volatile but potentially very rewarding. Its simplicity, combined with the innovative mechanic, made it an instant classic. The Legacy of Bonanza is that it set the standard for all future MegaWays games and remains a fan favorite.

Let's imagine that we want to create a mechanic close to Megaways and thus extract its beneficial properties, but without having to buy the license for this mechanic. In fact, Cayetano's design team have tried to do something similar before with the game **Classy Vegas** (without pots) released in 2018-05-15 (2 years after the first Megaways game). Here is what they did:



Did they manage to extract the Megaways beneficial properties? How do you think?

One of the most important features of the Megaways basic payout mechanic is related to volatility. It can explode with a big win stake multiplier at any time even if you are not in Free Spins bonus and this is due to the variability that many ways can inject. In how many ways can a player be paid in Classy Vegas? Well, lets calculate them. The calculation of all ways to win can be seen in the table on the right. We can see that **1,098 << 117,649** and we can conclude that the design team wasn't even close to replicating the mechanics of Megaways and certainly didn't manage to extract its beneficial properties. Still, they managed to do something different and innovative, but too complicated and failed to solve the task they set themselves. Admirations for the attempt, though. Once again simplicity proves to be more difficult than complexity and certainly much more profitable than it.

Classy Vegas: f(x, y) = f(x-1, y-1) + f(x-1, y+1)

0	0	0	0	0	0	0	0	0	0
1	1	2	3	6	10	19	33	61	108
1	2	3	6	10	19	33	61	108	197
1	2	4	7	13	23	42	75	136	244
1	2	4	7	13	23	42	75	136	244
1	2	3	6	10	19	33	61	108	197
1	1	2	3	6	10	19	33	61	108
0	0	0	0	0	0	0	0	0	0
Total ways to win:					1098				

So the first slot game that we explore was the Bonanza Megaways, which carried within itself the brilliant idea of its mechanics combining WAYS with WAYS MULTIPLUERS and calling it MEGAWAYS.

The second slot game that we explore was Classy Vegas, which we proved that was not very successful in terms of replication and leading to a completely different result, which is unfortunately too sophisticated and almost impossible to configure well, due to its synthetic mechanics.

Now lets explore the third slot game which does more than well in its attempt to inherit the winning characteristics of the Megaways mechanics in a natural way! This game is presented below and is a typical example of a good and simple idea.

So here it is, **Napoleon 2 Fat Stacks** slot developed by Blueprint Gaming and released in 2023-11-23. So the particular Napoleonic War theme game has a basic WAYS payout strategy with a screen 6 x 4 and respectively 4,096 ways to win. What's interesting here is what feature they managed to add to the regular Ways mechanics so that they get the beneficial properties of Megaways games without having to pay for a license and at the same time build on the basic mechanics in a natural and easily understandable way.

The idea is simple and it hides in the so called Fat Stacks:

Fat Stacks

If you land two or three identical single symbols on the reels, they can consolidate into double or triple symbols (Fat Stacks), creating even more ways to win at these slots. **Any symbol can be subject to the Fat Stacks feature, with the exception of wilds and scatters.**

You can check the game feeling here: [Napoleon 2 Fat Stacks DEMO LINK](#).

Without the other features, this game would have the potential to pass even Pennsylvania regulations, as only multiples of 2 and 3 and their combinations should be tracked. But anyway, the combination requirement was dropped with the regulation update from PGCB 2024-11-12. What we want to emphasize here is the original and simple idea that achieves the goals described above.